ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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C02-2

EXPERIMENT 7

Aim:

To determine and plot the capacitance of an isolated sphere and a concentric spherical capacitor for different radii and dielectric constants using MATLAB.

Test Case (i) – Capacitance of an Isolated Sphere

Determine the capacitance for a specified radius and relative permittivity.

MATLAB Program:

```
clc;
clear;
epsilon0 = 8.854e-12;
a = input('Enter radius of isolated sphere (m): ');
er = input('Enter relative permittivity: ');
epsilon = epsilon0*er;
C = 4*pi*epsilon*a;
disp('Capacitance of isolated sphere (F) =');
disp(C);
```

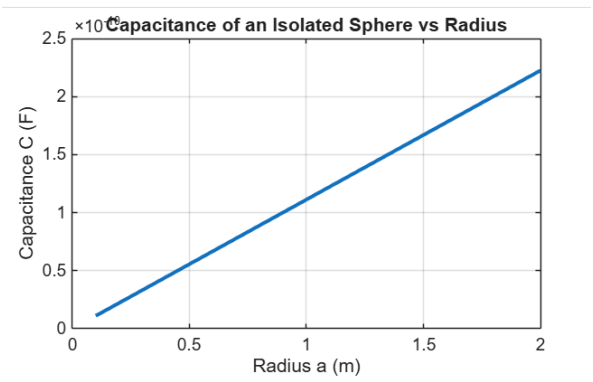
OUTPUT:

```
Radius of sphere = 0.50 m
Relative permittivity = 2.00
Capacitance of isolated sphere = 1.112626e-10 F
```

Test Case (ii) – Variation of Isolated-Sphere Capacitance with Radius

```
clc;
clear;
epsilon0 = 8.854e-12;
er = 1;
a = linspace(0.1,2,100);
C = 4*pi*epsilon0*er.*a;
plot(a,C,'LineWidth',2); grid on;
xlabel('Radius a (m)'); ylabel('Capacitance C (F)');
title('Capacitance of an Isolated Sphere vs Radius');
```

OUTPUT:



Test Case (iii) – Capacitance of a Concentric Spherical Capacitor

Determine the capacitance for specified inner radius a, outer radius b and relative permittivity εr.

```
clc;
clear;
epsilon0 = 8.854e-12;
a = input('Enter inner radius a (m): ');
b = input('Enter outer radius b (m): ');
er = input('Enter relative permittivity: ');
epsilon = epsilon0*er;
C = 4*pi*epsilon*a*b/(b-a);
disp('Capacitance of concentric spherical capacitor (F) =');
disp(C);
```

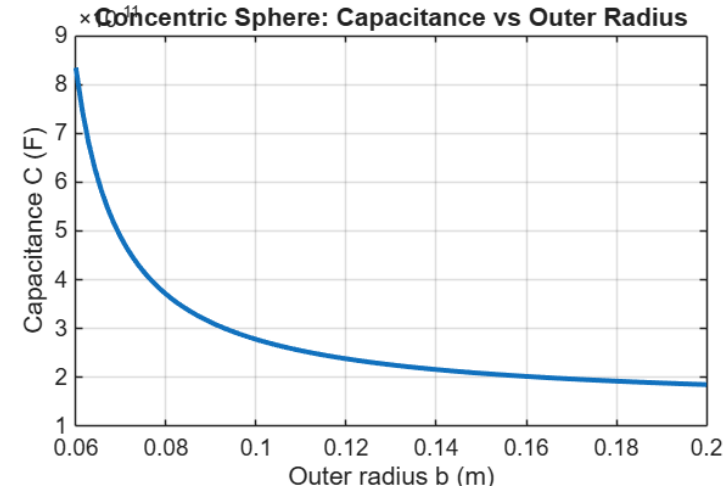
OUTPUT:

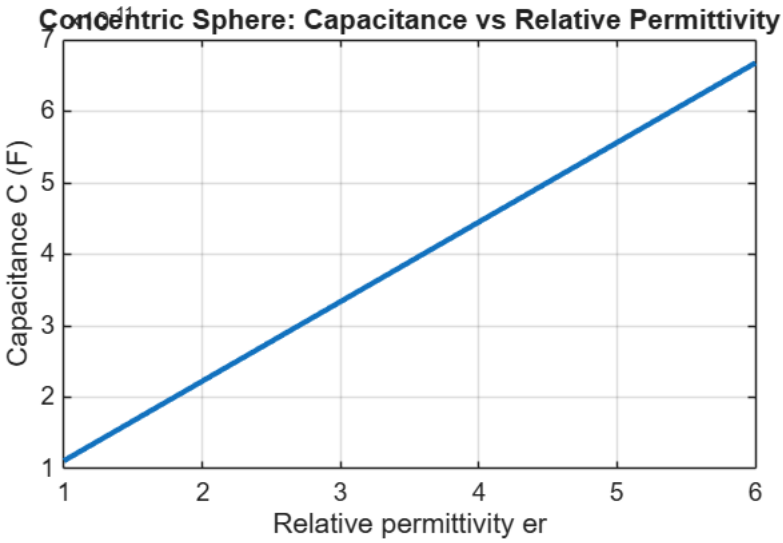
```
Inner radius a = 0.050 m
Outer radius b = 0.100 m
Relative permittivity = 2.00
Capacitance of concentric spherical capacitor = 2.225253e-11 F
```

Test Case (iv) – Effect of Outer Radius and Relative Permittivity

```
clc;
 clear;
epsilon0 = 8.854e-12;
a = 0.05;
er = 2.5;
b = linspace(0.06,0.20,100);
C1 = 4*pi*epsilon0*er.*a.*b./(b-a);
figure; plot(b,C1,'LineWidth',2);
grid on;
xlabel('Outer radius b (m)');
 ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Outer Radius');
er = linspace(1,6,100);
b = 0.10;
C2 = 4*pi*epsilon0.*er.*a.*b./(b-a);
figure; plot(er,C2,'LineWidth',2);
grid on;
xlabel('Relative permittivity er');
ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Relative Permittivity');
```

OUTPUT:


Concentric Sphere: Capacitance vs Outer Radius
×10^-11
Capacitance C (F)
Outer radius b (m)
9
8
7
6
5
4
3
2
1
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2


Concentric Sphere: Capacitance vs Relative Permittivity
×10^-11
Capacitance C (F)
Relative permittivity er
7
6
5
4
3
2
1
1
2
3
4
5
6

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weighting**	**Marks given**	**Marks obtained**
Define spherical geometry and dielectric	Define spherical geometry and dielectric	Correctly identifies isolated-sphere radius or inner/outer radii and medium permittivity.	Geometry is correct with minor omission.	Some dimensions or material properties are unclear.	Uses invalid geometry or material inputs.	25	2.5	
Implement spherical capacitance equations	Implement spherical capacitance equations	Applies the appropriate isolated/concentric sphere formulas correctly.	Formulas are mostly correct with minor errors.	Partial calculations require guidance.	Uses incorrect equations.	25	2.5	
Analyze radius and permittivity effects	Analyze radius and permittivity effects	Compares cases and validates the expected capacitance trends against theory.	Analysis is mostly correct with minor omissions.	Comparisons or validation are incomplete.	No valid trend analysis is shown.	30	3.0	
Present results and conclusions	Present results and conclusions	Reports geometry, units, calculations and conclusions clearly.	Report is clear with minor gaps.	Report lacks some key information.	Results are unclear or missing.	20	2.0	
TOTAL						100	10	